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SYNCHRONIZATION MECHANISM AND FOLDING HINGE

Abstract

Provided are a synchronization mechanism and a folding hinge. In the synchronization mechanism, a base includes a spindle seat and a stop rack, and one first rotation portion is provided on each of two sides of the spindle seat along the width direction; the stop rack is detachably connected to the spindle seat, and a second rotation portion is provided on each of two sides of the stop rack along the width direction; two swing arms are rotatably disposed on two first rotation portions respectively; at least two slide rail arms are synchronously and rotatably disposed on two second rotation portions respectively, and each slide rail arm is provided with a slide groove; one end of a connecting rod shaft is connected to a respective swing arm while the other end of the connecting rod shaft is in a sliding fit with the slide groove.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application claims priority to Chinese Patent Application No. 202410259058.1 filed with the China National Intellectual Property Administration (CNIPA) on Mar. 7, 2024, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of folding screens and, in particular, a synchronization mechanism and a folding hinge.

BACKGROUND

[0003] A flexible screen is becoming thinner, which requires two sets of torsion structures to meet a torsion demand. To prevent the two sets of torsion structures from moving away from each other, a slide connection is generally implemented by using a housing connecting rod and two slide rail arms, that is, the housing connecting rod limits the positions of the two slide rail arms in the length direction so that the two torsion structures are prevented from moving away from each other. However, the manner in which the housing connecting rod is slidably connected to the two slide rail arms is prone to a synchronization drop in a rotation process, thereby causing a torsion between the two slide rail arms and the housing connecting rod. As a result, one of the two slide rail arms is easily detached from the housing connecting rod. For ease of installation, a driving shaft is generally used for connecting a base and the two sets of torsion assemblies in series. Due to a large number of parts connected in series, machining precision between the parts is high, thereby increasing machining cost.

[0004] Therefore, a synchronization mechanism must be studied urgently to solve the problem of an easy detachment of a slide rail arm and a housing connecting rod in the synchronization mechanism and the high machining cost.

SUMMARY

[0005] An object of the present disclosure is to provide a synchronization mechanism and a folding hinge to solve the problems of easy detachment of a slide rail arm and a housing connecting rod in the synchronization mechanism and the high machining cost in the related art.

[0006] To achieve the preceding object, the present disclosure provides the technical solutions below.

[0007] A synchronization mechanism is provided. The synchronization mechanism includes a base, swing arms, slide rail arms and a connecting rod shaft.

[0008] The base includes a spindle seat and a stop rack.

[0009] A first rotation portion is provided on each of the two sides of the spindle seat along a width direction.

[0010] The stop rack is detachably connected to the spindle seat, and a second rotation portion is provided on each of the two sides of the stop rack along the width direction.

[0011] The swing arms are rotatably disposed on two first rotation portions respectively.

[0012] At least two slide rail arms are synchronously and rotatably disposed on two second rotation portions respectively, and each slide rail arm is provided with a slide groove.

[0013] One end of the connecting rod shaft is connected to a swing arm, and the other end of the

connecting rod shaft is in a sliding fit with the slide groove, and the swing arm is driven to rotate through the slide groove and the connecting rod shaft in a rotation process of a respective slide rail arm.

[0014] In some embodiments, one of the stop rack or the spindle seat is provided with an overlapping groove, and the other one of the stop rack or the spindle seat is provided with an inserting protrusion, and the inserting protrusion is inserted into the overlapping groove.

[0015] In some embodiments, the overlapping groove is disposed on the spindle seat, the overlapping groove passes through the spindle seat along a direction facing the stop rack, the overlapping groove is a stepped groove, and two first limit portions opposite to each other are formed at an opening of the stepped groove.

[0016] The inserting protrusion is disposed on the stop rack, the inserting protrusion includes an inserting body extending along a length direction and second limit portions disposed at an end of the inserting body and extending along the width direction, a width of the inserting protrusion at the second limit portions is greater than a width of the overlapping groove at the two first limit portions, and the second limit portions are disposed on one side facing away from the opening of the overlapping groove.

[0017] In some embodiments, the swing arm includes a swing body and a connecting portion connected to one end of the swing body, the connecting portion is in a rotation fit with the first rotation portion, the swing body is provided with a swing hole, and the connecting rod shaft passes through and is secured within the swing hole.

[0018] In some embodiments, the swing body is provided with an arc-shaped groove passing through the swing body along the length direction, each swing arm further includes a cover plate covering the arc-shaped groove, the swing hole is formed by an enclosure between the cover plate and the arc-shaped groove, and the length of the cover plate is less than the length of the arc-shaped groove in the length direction.

[0019] In some embodiments, at least two connecting rod shafts are provided in one-to-one correspondence with the at least two slide rail arms, and each connecting rod shaft has a rod body and a stop portion disposed at an end of the rod body. The rod body passes through the swing hole, the outer diameter of the stop portion is greater than the outer diameter of the swing hole, and the stop portion is located on one side of the cover plate facing away from the slide rail arm.

[0020] In some embodiments, an end surface of the stop portion is fan-shaped and disposed around an outer periphery of the rod body, and two ends of the stop portion around its axis abut against the swing body and are located on two sides of the arc-shaped groove, respectively.

[0021] In some embodiments, the rod body includes a first shaft and a second shaft, the diameter of the first shaft is greater than the diameter of the second shaft and greater than the width of the slide groove, the stop portion is disposed at an end of the first shaft facing away from the second shaft, the second shaft is slidably disposed in the slide groove, and the first shaft is in an interference fit with the swing hole or is bonded to the swing hole.

[0022] In some embodiments, the at least two slide rail arms rotate between an initial position and an end position, and the slide groove is bent downward gradually along a direction facing away from the base.

[0023] In some embodiments, each of the at least two slide rail arms is provided with a slide protrusion, the slide protrusion is configured to be in a sliding fit with a slide rail groove of a housing connecting rod, and the slide protrusion is provided with a notch for avoidance of the connecting rod shaft.

[0024] In some embodiments, the stop rack includes a stop body and two hinged protrusions disposed on two sides of the stop body respectively, the second rotation portion includes a hinged hole passing through a respective one of the two hinged protrusions along the length direction, two hinged blocks opposite to each other are provided on one end of each of the at least two slide rail arms, a hinged groove is formed between the two hinged blocks, a rotation hole passes through

each of the two hinged blocks along the length direction, the two hinged protrusions are inserted into the hinged groove, and a driving shaft passes through the hinged hole and the rotation hole. [0025] In some embodiments, each hinged protrusion is provided with a first limit protrusion and a second limit protrusion, the first limit protrusion is arc-shaped and disposed around an outer periphery of the hinged hole, and the second limit protrusion is arc-shaped and disposed around an outer periphery of the rotation hole. The slide rail arm rotates between an initial position and an end position with respect to the stop rack, when the slide rail arm is at the initial position, one end of the first limit protrusion abuts against one end of the second limit protrusion, and when the slide rail arm is at the end position, the other end of the first limit protrusion abuts against the other end of the second limit protrusion.

[0026] In some embodiments, the driving shaft includes a shaft body and a shaft limit portion, the outer diameter of the shaft body is less than the outer diameter of the shaft limit portion, the shaft body passes through the two hinged blocks and the rotation hole, the shaft limit portion is located on a side of the respective slide rail arm facing the spindle seat, and the spindle seat is provided with an avoidance notch for accommodating the shaft limit portion.

[0027] In some embodiments, two torsion assemblies are also included, each torsion assembly includes two intermediate gears engaging with each other and two main gears disposed on outer sides of the two intermediate gears respectively, two slide rail arms on the same stop rack are connected to the two main gears respectively and rotate around the axes of the two main gears, respectively, the two main gears and the two slide rail arms are relatively secured in a circumferential direction, the two intermediate gears are rotatably disposed on the stop rack, the two intermediate gears engage with the two main gears respectively, and the driving shaft passes through a gear hole of the two main gears.

[0028] In some embodiments, the main gear is provided with a limit protruding block located on a circumferential side of the gear hole, the hinged block is provided with a limit groove disposed on a circumferential side of the rotation hole, and the limit protruding block is inserted into the limit groove.

[0029] Two limit grooves are provided, one of the two limit grooves communicates with the rotation hole, the other one is spaced from the rotation hole, and two limit protruding blocks are provided and inserted into the two limit grooves, respectively.

[0030] A folding hinge is provided. The folding hinge includes door plates, housing connecting rods and the synchronization mechanism according to any preceding technical solution. Two housing connecting rods are located on two sides of the base along a width direction respectively, are in a sliding fit with slide rail arms respectively and are hinged with swing arms respectively, and two door plates are located on the two sides of the base along the width direction respectively, are in a rotation fit with two housing connecting rods respectively and are in a rotation fit with two connecting rod shafts respectively.

[0031] In some embodiments, a rotation groove is provided on a side of each of the two housing connecting rods facing a respective one of the two swing arms, one first connecting hole is provided on each of the opposite sidewalls of the rotation groove, a rotation protrusion is provided on a side of the respective swing arm facing the housing connecting rod, a second connecting hole passes through the rotation protrusion along a length direction, the rotation protrusion is inserted into the rotation groove, a connecting shaft passes through the second connecting hole, and two ends of the connecting shaft are located within two first connecting holes respectively.

[0032] In some embodiments, the folding hinge further includes a median plate. The spindle seat is provided with an overlapping groove, the stop rack is provided with an inserting protrusion, the inserting protrusion is inserted into the overlapping groove, the overlapping groove passes through the spindle seat along a thickness direction, and the median plate is soldered onto the spindle seat and covers the overlapping groove.

[0033] The present disclosure has the beneficial effects below.

[0034] The present disclosure provides a synchronization mechanism and a folding hinge. The synchronization mechanism includes the base, the swing arms, the slide rail arms and the connecting rod shaft. The base includes the spindle seat and the stop rack that are separate and detachably connected. Therefore, when the synchronization mechanism and torsion assemblies are assembled, the torsion assemblies are connected to the stop rack first, and then the stop rack is connected to the spindle seat so that a driving shaft can be prevented from passing through the spindle seat, and the two torsion assemblies can also be prevented from sharing the same one driving shaft, thereby reducing the demand for the length of the driving shaft, reducing the requirement for the machining precision and lowering the machining cost. In addition, the swing arms are rotatably connected to the spindle seat, the slide rail arms are rotatably connected to the stop rack, the swing arms are connected to the at least two slide rail arms through the connecting rod shaft so that in the rotation process, the slide rail arms are limited by the swing arms and the connecting rod shaft, the consistency of rotation angles of the slide rail arms and the housing connecting rods can be ensured, and the problem of easy detachment of the slide rail arms and the housing connecting rods can be solved.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0035] FIG. 1 is a view illustrating the structure of a folding hinge taken from a first perspective according to an embodiment of the present disclosure.

[0036] FIG. 2 is a view illustrating the structure of a folding hinge taken from a second perspective according to an embodiment of the present disclosure.

[0037] FIG. 3 is a cross-sectional view taken along A-A in FIG. 2.

[0038] FIG. 4 is a view illustrating the partial structure of a folding hinge taken from a third perspective according to an embodiment of the present disclosure.

[0039] FIG. 5 is a view illustrating the structure of a synchronization mechanism taken from a first perspective according to an embodiment of the present disclosure.

[0040] FIG. 6 is a view illustrating the structure of a synchronization mechanism taken from a second perspective according to an embodiment of the present disclosure.

[0041] FIG. 7 is a view illustrating the structure of a synchronization mechanism taken from a third perspective according to an embodiment of the present disclosure.

[0042] FIG. 8 is a view illustrating the structure of a first limit protrusion and a second limit protrusion according to an embodiment of the present disclosure.

[0043] FIG. 9 is a view illustrating the structure of a slide rail arm according to an embodiment of the present disclosure.

[0044] FIG. 10 is a view illustrating the structure of a spindle seat according to an embodiment of the present disclosure.

[0045] FIG. 11 is a view illustrating the structure of a stop rack according to an embodiment of the present disclosure.

[0046] FIG. 12 is a view illustrating the structure of a main gear according to an embodiment of the present disclosure.

[0047] FIG. 13 is a view illustrating the partial structure of a folding hinge taken from a fourth perspective according to an embodiment of the present disclosure.

[0048] FIG. 14 is an enlarged view at J in FIG. 13.

REFERENCE LIST

[0049] **100** base

[0050] **110** spindle seat

[0051] **111** overlapping groove

[0052] **112** first rotation portion
[0053] **113** first limit portion
[0054] **120** stop rack
[0055] **121** stop body
[0056] **122** hinged protrusion
[0057] **1221** second rotation protrusion
[0058] **1222** first limit protrusion
[0059] **123** inserting protrusion
[0060] **1231** second limit portion
[0061] **200** swing arm
[0062] **210** swing body
[0063] **211** arc-shaped groove
[0064] **212** cover plate
[0065] **220** connecting portion
[0066] **230** swing hole
[0067] **240** rotation protrusion
[0068] **300** slide rail arm
[0069] **310** hinged block
[0070] **311** hinged groove
[0071] **312** second limit protrusion
[0072] **313** limit groove
[0073] **314** rotation hole
[0074] **320** slide protrusion
[0075] **321** notch
[0076] **330** slide groove
[0077] **400** connecting rod shaft
[0078] **410** rod body
[0079] **420** stop portion
[0080] **500** driving shaft
[0081] **510** shaft body
[0082] **520** shaft limit portion
[0083] **610** door plate
[0084] **611** door hole
[0085] **620** median plate
[0086] **700** housing connecting rod
[0087] **710** slide rail groove
[0088] **720** rotation groove
[0089] **730** connecting shaft
[0090] **800** torsion assembly
[0091] **810** main gear
[0092] **811** limit protruding block
[0093] **812** gear hole
[0094] **820** intermediate gear

DETAILED DESCRIPTION

[0095] The technical solutions of the present disclosure are described clearly and completely hereinafter in conjunction with drawings. Apparently, the described embodiments are part, not all, of embodiments of the present disclosure. Based on the embodiments of the present disclosure, all other embodiments obtained by those of ordinary skill in the art without creative work are within the scope of the present disclosure.

[0096] In the description of the present disclosure, it is to be noted that orientations or position

relations indicated by terms such as “center”, “upper”, “lower”, “left”, “right”, “vertical”, “horizontal”, “in” and “out” are based on the drawings. These orientations or position relations are intended only to facilitate and simplify the description of the present disclosure and not to indicate or imply that a device or element referred to must have such particular orientations or must be configured or operated in such particular orientations. Thus, these orientations or position relations are not to be construed as limiting the present disclosure. In addition, terms such as “first” and “second” are used only for the purpose of description and are not to be construed as indicating or implying relative importance. Terms “first position” and “second position” are two different positions. Moreover, when a first feature is described as “on”, “above”, or “over” a second feature, the first feature is right on, above, or over the second feature, the first feature is obliquely on, above, or over the second feature, or the first feature is simply at a higher level than the second feature. When the first feature is described as “under”, “below”, or “underneath” the second feature, the first feature is right under, below, or underneath the second feature, the first feature is obliquely under, below, or underneath the second feature, or the first feature is simply at a lower level than the second feature.

[0097] In the description of the present disclosure, it is to be noted that unless otherwise expressly specified and limited, the terms “mounted”, “connected to each other”, or “connected” should be construed in a broad sense as securely connected, detachably connected, or integrally connected; mechanically connected or electrically connected; directly connected to each other or indirectly connected to each other via an intermediary; or interconnected between two components. For those of ordinary skill in the art, specific meanings of the preceding terms in the present disclosure may be construed based on specific situations.

[0098] Embodiments of the present disclosure are described in detail below, and examples of the embodiments are illustrated in the drawings, where the same or similar reference numerals indicate the same or similar elements or elements having the same or similar functions. The embodiments described below with reference to the drawings are solely exemplary and intended to explain the present disclosure, and are not to be construed as limiting the present disclosure.

[0099] As shown in FIGS. **1** to **14**, this embodiment provides a synchronization mechanism. The synchronization mechanism includes a base **100**, swing arms **200**, slide rail arms **300** and a connecting rod shaft **400**. The base **100** includes a spindle seat **110** and a stop rack **120**. A first rotation portion **112** is provided on each side of the spindle seat **110** along the width direction. The stop rack **120** is detachably connected to the spindle seat **110**. A second rotation portion **1221** is provided on each side of the stop rack **120** along the width direction. Two swing arms **200** are rotatably disposed on two first rotation portions **112** respectively. At least two slide rail arms **300** are synchronously and rotatably disposed on the two second rotation portions **1221** respectively. Each slide rail arm **300** is provided with a slide groove **330**. One end of the connecting rod shaft **400** is connected to a respective swing arm **200** while the other end of the connecting rod shaft **400** is in a sliding fit with the slide groove **330**. The swing arm **200** is driven to rotate through the slide groove **330** and the connecting rod shaft **400** in a rotation process of a respective slide rail arm **300**.

[0100] Two torsion assemblies **800** are provided and located on two sides of the base **100** along the length direction respectively. Two stop racks **120** are provided and located on two sides of the spindle seat **110** along the length direction respectively. The two torsion assemblies **800** are connected to the two stop racks **120** respectively. Four slide rail arms **300** are provided. Two slide rail arms **300** are rotatably connected to each side of the base **100** along the width direction. The two slide rail arms **300** located on the same side of the base **100** are located on two sides of a respective swing arm **200** along the length direction respectively. Each swing arm **200** is connected to two slide rail arms **300** through the connecting rod shaft **400**. In other words, as shown in FIGS. **6** and **7**, two sides of the same swing arm **200** along the length direction are each secured with one connecting rod shaft **400**, and each connecting rod shaft **400** is in a sliding fit with a respective slide rail arm **300**. For ease of understanding, in this embodiment, the length direction is defined as

the axis direction of the connecting rod shaft **400**, and the width direction is defined as the direction of an interval between two slide rail arms **300** on the same stop rack **120**.

[0101] With the preceding arrangement, when the synchronization mechanism and the two torsion assemblies **800** are assembled, the two torsion assemblies **800** are connected to the two stop racks **120** respectively, and then the two stop racks **120** are connected to the spindle seat **110** so that a driving shaft **500** can be prevented from passing through the spindle seat **110**, and the two torsion assemblies **800** can also be prevented from sharing the same driving shaft **500**, thereby reducing the demand for the length of the driving shaft **500**, reducing the requirement for the machining precision and lowering the machining cost. In addition, the swing arms **200** are rotatably connected to the spindle seat **110**, the slide rail arms **300** are rotatably connected to the stop racks **120**, and the swing arms **200** are connected to the slide rail arms **300** through the connecting rod shafts **400** respectively so that the slide rail arms **300** are limited by the swing arms **200** and the connecting rod shafts **400** in the rotation process, the consistency of rotation angles can be ensured, and the problem caused by easy detachment of the slide rail arms **300** and a housing connecting rod **700** can be solved.

[0102] The arrangement of the swing arms **200** plays a role in connecting two slide rail arms **300** so that the rotations of the two slide rail arms **300** can remain in synchronization, thereby ensuring the consistency in the rotation angle of each slide rail arm **300** and the housing connecting rod **700** and avoiding a detachment of the slide rail arms **300** and the housing connecting rod **700**.

[0103] In some embodiments, one of the stop rack **120** or the spindle seat **110** is provided with an overlapping groove **111** while the other one of the stop rack **120** or the spindle seat **110** is provided with an inserting protrusion **123**, and the inserting protrusion **123** is inserted into the overlapping groove **111**. The insertion is simple and can quickly connect the stop rack **120** to the spindle seat **110**.

[0104] The overlapping groove **111** is disposed on the spindle seat **110**, the overlapping groove **111** passes through the spindle seat **110** along the direction facing the stop rack **120**, the overlapping groove **111** is a stepped groove, and two first limit portions **113** opposite to each other are formed at an opening of the stepped groove. The inserting protrusion **123** is disposed on the stop rack **120**, the inserting protrusion **123** includes an inserting body extending along the length direction and second limit portions **1231** disposed at an end of the inserting body and extending along the width direction, a width of the inserting protrusion **123** at the second limit portions **1231** is greater than a width of the overlapping groove **111** at the two first limit portions **113**, and the second limit portions **1231** are disposed on a side facing away from the opening of the overlapping groove **111**. With the help of the preceding structure, the stop rack **120** and the spindle seat **110** are interconnected so that the stop rack **120** can be effectively prevented from moving toward the direction away from the spindle seat **110**, and the stability of the connection between the stop rack **120** and the spindle seat **110** can be ensured. In other embodiments, the stop rack **120** and the spindle seat **110** may also be threaded through screws. The axis of each screw is perpendicular to the length direction.

[0105] In some embodiments, the swing arm **200** includes a swing body **210** and a connecting portion **220** connected to one end of the swing body **210**, the connecting portion **220** is in a rotation fit with the first rotation portion **112**, the swing body **210** is provided with a swing hole **230**, and the connecting rod shaft **400** passes through and is secured in the swing hole **230**. With the preceding arrangement, the swing arms **200** are connected to the spindle seat **110**, and the connecting rod shaft **400** passes through the swing hole **230** so that while being secured, the connecting rod shaft **400** can be prevented from passing through the entire swing arm **200**, effectively reducing the length of the connecting rod shaft **400** and thereby reducing the coaxiality requirement and machining cost of the connecting rod shaft **400**. In the length direction, the length of the swing hole **230** is roughly one-fifth of the length of the swing arm **200**. In the width direction, two swing holes **230** are spaced apart on the swing arm **200** and are configured to secure

the two connecting rod shafts **400** respectively.

[0106] The swing body **210** is provided with an arc-shaped groove **211**. The arc-shaped groove **211** penetrates the swing body **210** along the length direction. The swing arm **200** further includes a cover plate **212** covering the arc-shaped groove **211**. The swing hole **230** is formed by an enclosure between the cover plate **212** and the arc-shaped groove **211**. In the length direction, the length of the cover plate **212** is less than the length of the arc-shaped groove **211**. The preceding arrangement limits the manner for forming the swing hole **230**. For one thing, the material for the swing arm **200** can be reduced effectively, lowering the cost, and for another thing, with the semi-open hole structure, the overall thickness of the swing arm **200** can be reduced effectively with the help of the arc-shaped groove **211** on the basis of providing the swing hole **230**, thereby facilitating the reduction in the thickness of a folding hinge to which the synchronization mechanism is applied and further being more adaptive to thinner and thinner folding screen products.

[0107] At least two connecting rod shafts **400** are provided. The at least two connecting rod shafts **400** are in one-to-one correspondence with the at least two slide rail arms **300**. Each connecting rod shaft **400** has a rod body **410** and a stop portion **420** disposed at an end of the rod body **410**. The rod body **410** passes through the swing hole **230**. The outer diameter of the stop portion **420** is greater than the outer diameter of the swing hole **230**. The stop portion **420** is located on one side of the cover plate **212** facing away from the respective slide rail arm **300**. The preceding arrangement facilitates the synchronous rotation of each slide rail arm **300** with the housing connecting rod **700**. In addition, the arrangement of the stop portion **420** limits the displacement of the connecting rod shaft **400** in the length direction, avoids the movement of the connecting rod shaft **400** in the direction facing the slide rail arm **300** and also plays a limit role in the installation of the connecting rod shaft **400**. During installation, clear feedback of installation in place can be obtained as long as the stop portion **420** abuts against the end of the swing hole **230**.

[0108] In some embodiments, the end surface of the stop portion **420** is fan-shaped and disposed around the outer periphery of the rod body **410**, and two ends of the stop portion **420** around its axis abut against the swing body **210** and are located on two sides of the arc-shaped groove **211** respectively. The preceding arrangement achieves a circumferential limit of the connecting rod shaft **400** and avoids the rotation of the connecting rod shaft **400** so that wear on the swing hole **230** can be reduced, facilitating the reduction in the requirement for the wall thickness of the swing hole **230** and reducing the requirement for the overall thickness of the swing arm **200**.

[0109] The rod body **410** includes a first shaft and a second shaft. The diameter of the first shaft is greater than the diameter of the second shaft and greater than the width of the slide groove **330**. The stop portion **420** is disposed at the end of the first shaft facing away from the second shaft. The second shaft is slidably disposed in the slide groove **330**. The first shaft is in an interference fit with the swing hole **230** or is bonded to the swing hole **230**. The preceding arrangement facilitates the reduction in the width of the slide groove **330** so that the overall thickness of the slide rail arm **300** can be reduced and further the thickness of the synchronization mechanism and the thickness of the folding hinge can be reduced. When the first shaft is bonded to the swing hole **230**, for one thing, more glue can be accommodated so that the connecting rod shaft **400** can be firmly bonded to the swing body **210**; and for another thing, glue can also be prevented from overflowing.

[0110] The slide rail arm **300** rotates between an initial position and an end position. The slide groove **330** is bent downward gradually along the direction facing away from the base **100**. With the preceding arrangement, when the slide rail arm **300** is located at the end position, the rotation angle of the swing arm **200** is greater than the rotation angle of the slide rail arm **300**, which facilitates a droplet shape to be formed during a folding process of the screen, thereby avoiding a noticeable crease at a bent point of the screen and improving durability of the screen.

[0111] To ensure that the screen forms a droplet-shaped structure, a certain requirement for the curvature of the slide groove **330** is provided. In addition, considering the development trend of decreasing the thickness of an electronic device, improvements are made to the structure of the

slide rail arm **300**. In some embodiments, the slide rail arm **300** is provided with a slide protrusion **320**. The slide protrusion **320** is configured to be in a sliding fit with a slide rail groove **710** of the housing connecting rod **700**. The slide protrusion **320** is provided with a notch **321** for avoidance of the connecting rod shaft **400**. The preceding arrangement can not only meet the slide requirement for the slide rail arm **300** and the housing connecting shaft **700** but also meet the curvature requirement for the slide groove **330** and effectively reduce the thickness of the slide rail arm **300**. The notch **321** is located at one end of the slide protrusion **320** facing away from the spindle seat **110**. It is to be noted that when the screen is flattened, the slide protrusion **320** of the slide rail arm **300** is located in the slide rail groove **710**, and the notch **321** is also located in the slide rail groove **710**; and when the screen is bent, the slide rail arm **300** is displaced relative to the housing connecting shaft **700**, and a part (the part bound by the notch **321** and facing the hinged block **310**) of the slide protrusion **320** and the corresponding part of the notch **321** move out of the slide rail groove **710** so that the connecting rod shaft **400** can move to the notch **321** smoothly. In addition, after the notch **321** moves out of the slide rail groove **710**, the other part (the part bound by the notch **321** and away from the hinged block **310**) of the slide protrusion **320** is still in the slide rail groove **710** to ensure the reliability of the connection between the slide rail arm **300** and the housing connecting rod **700**.

[0112] Further, an avoidance portion is provided on one side of the housing connecting rod **700** facing the connecting rod shaft **400** for avoidance of the notch **321** so that the notch **321** can move out of the slide rail groove **710** as early as possible. In other words, that is, among two slide rail grooves **710** that are on the housing connecting rod **700** and correspond to the same slide rail arm **300**, the length of the outer slide rail groove **710** is greater than the length of the inner slide rail groove **710**.

[0113] The stop rack **120** includes a stop body **121** and two hinged protrusions **122** disposed on two sides of the stop body **121** respectively. The second rotation portion **1221** includes a hinged hole passing through a respective hinged protrusion **122** along the length direction. One end of the slide rail arm **300** is provided with two hinged blocks **310** opposite to each other. A hinged groove **311** is formed between the two hinged blocks **310**. Each hinged block **310** is provided with a rotation hole **314** penetrating the hinged block **310** along the length direction. The hinged protrusion **122** is inserted into the hinged groove **311**. A driving shaft **500** passes through the hinged hole and the rotation hole **314**. The preceding arrangement achieves the rotation connection between the stop rack **120** and the slide rail arm **300** and can effectively limit the movement of the slide rail arm **300** with respect to the stop rack **120** in the length direction so that the stability of the connection between the slide rail arm **300** and the stop rack **120** can be improved.

[0114] The driving shaft **500** includes a shaft body **510** and a shaft limit portion **520**. The outer diameter of the shaft body **510** is less than the outer diameter of the shaft limit portion **520**. The shaft body **510** passes through the hinged blocks **310** and the rotation hole **314**. The shaft limit portion **520** is located on one side of the slide rail arm **300** facing the spindle seat **110**. The spindle seat **110** is provided with an avoidance notch for accommodating the shaft limit portion **520**.

[0115] Each hinged protrusion **122** is provided with a first limit protrusion **1222**. The first limit protrusion **1222** is arc-shaped and disposed around the outer periphery of the hinged hole. Each hinged block **310** is provided with a second limit protrusion **312**. The second limit protrusion **312** is arc-shaped and disposed around the outer periphery of the rotation hole **314**. The slide rail arm **300** rotates between the initial position and the end position with respect to the stop rack **120**, when the slide rail arm **300** is at the initial position, one end of the first limit protrusion **1222** abuts against one end of the second limit protrusion **312**, and when the slide rail arm **300** is at the end position, the other end of the first limit protrusion **1222** abuts against the other end of the second limit protrusion **312**. The preceding arrangement achieves a limitation on the rotation extreme position of the slide rail arm **300**. Due to the arrangement of the second limit protrusion **312**, the thickness of the hinged block **310** is increased so that the structural strength of the hinged block **310** can be

increased, thereby facilitating the improvement in the reliability of the slide rail arm **300**.

[0116] Each torsion assembly **800** includes two intermediate gears **820** engaging with each other and two main gears **810** disposed on outer sides of the two intermediate gears **820** respectively. Two slide rail arms **300** on the same stop rack **120** are connected to the two main gears **810** respectively and rotate around the axes of the two main gears **810** respectively. The two main gears **810** and the slide rail arms **300** are relatively secured in the circumferential direction. The two intermediate gears **820** are rotatably disposed on the stop rack **120**. The two intermediate gears **820** engage with the two main gears **810** respectively. The driving shaft **500** passes through a gear hole **812** of the main gear **810**. The main gear **810** is provided with a limit protruding block **811** located on a circumferential side of the gear hole **812**. The hinged block **310** is provided with a limit groove **313** disposed on a circumferential side of the rotation hole **314**. The limit protruding block **811** is inserted into the limit groove **313**. Two limit grooves **313** are provided, one of the two limit grooves **313** communicates with the rotation hole **314** while the other one of the two limit grooves **313** is spaced from the rotation hole **314**. Two limit protruding blocks **811** are provided and inserted into the two limit grooves **313** respectively.

[0117] Two limit grooves **313** are provided on the back of each hinged block **310**. The preceding arrangement may allow the slide rail arm **300** to be used at any position, that is, the slide rail arm **300** at each position may be replaced with each other, so that the number of mode openings can be reduced effectively, the cost can be lowered, and the installation difficulty can be reduced.

[0118] This embodiment further provides a folding hinge. The folding hinge includes door plates **610**, housing connecting rods **700** and the synchronization mechanism in the preceding embodiment. Two housing connecting rods **700** are located on the two sides of the base **100** along the width direction respectively, are in a sliding fit with the slide rail arms **300** respectively and are hinged with a respective swing arm **200**. Two door plates **610** are located on the two sides of the base **100** along the width direction respectively, are in a rotation fit with the housing connecting rods **700** respectively and are in a rotation fit with connecting rod shafts **400** respectively.

[0119] The preceding arrangement prevents the driving shaft **500** from passing through the spindle seat **110** and also prevents the two torsion assemblies **800** from sharing the same driving shaft **500** so that the demand for the length of the driving shaft **500** can be reduced, the requirement for the machining precision can be reduced, and the machining cost can be lowered. In addition, the swing arms **200** are rotatably connected to the spindle seat **110**, the slide rail arms **300** are rotatably connected to the stop rack **120**, and the swing arms **200** are connected to the slide rail arms **300** through the connecting rod shafts **400** so that in the rotation process, the slide rail arms **300** are limited by the swing arms **200** and the connecting rod shafts **400**, the consistency in the rotation angle can be ensured, and the problem caused by easy detachment of the at least two slide rail arms **300** and the two housing connecting rods **700** can be solved. Finally, when the folding hinge is used, the connection relationship of the door plates **610** enables the door plates **610** to flip with respect to the housing connecting rods **700** respectively so that the screen can generate a droplet structure, avoiding creases at bent points of the screen.

[0120] Each door plate **610** is provided with a door hole **611**, and the connecting rod shaft **400** is configured to pass through the door hole **611** to drive the door plate **610** to rotate with respect to the respective housing connecting rod **700** so that the screen can be extruded to form a droplet shape.

[0121] In some embodiments, one side of the housing connecting rod **700** facing the swing arm **200** is provided with a rotation groove **720**, one first connecting hole is provided on each of the opposite sidewalls of the rotation groove **720**, a rotation protrusion **240** is provided on one side of the swing arm **200** facing the housing connecting rod **700**, the rotation protrusion **240** is provided with a second connecting hole penetrating the rotation protrusion **240** along the length direction, the rotation protrusion **240** is inserted into the rotation groove **720**, a connecting shaft **730** passes through the second connecting hole, and two ends of the connecting shaft **730** are located in two first connecting holes respectively. The preceding arrangement connects the housing connecting

rods **700** to the swing arms **200** and avoids a relative displacement between the housing connecting rods **700** and the swing arms **200** in the length direction so that the housing connecting rods **700** can be stably connected to the swing arms **200** respectively.

[0122] In some embodiments, the folding hinge further includes a median plate **620**, the spindle seat **110** is provided with the overlapping groove **111**, the stop rack **120** is provided with the inserting protrusion **123**, the inserting protrusion **123** is inserted into the overlapping groove **111**, the overlapping groove **111** penetrates the spindle seat **110** along the thickness direction, and the median plate **620** is soldered onto the spindle seat **110** and covers the overlapping groove **111**. With the help of the preceding structure, for one thing, the median plate **620** is connected to the spindle seat **110**; and for another thing, the median plate **620** seals the overlapping groove **111**. Therefore, during assembly, before the overlapping groove **111** is sealed, the stop rack **120** may be in an inserting fit with the spindle seat **110** easily, and after the median plate **620** is connected, the stop rack **120** is secured to the spindle seat **110** in an all-round manner.

[0123] Apparently, the preceding embodiments of the present disclosure are only illustrative examples of the present disclosure and are not intended to limit the implementations of the present disclosure. For those of ordinary skill in the art, changes or alterations in other different forms may also be made based on the preceding description. All examples cannot be and do not need to be exhausted herein. Any modifications, equivalent substitutions and improvements made within the spirit and principle of the present disclosure fall within the scope of the claims of the present disclosure.

Claims

1. A synchronization mechanism, comprising: a base, wherein the base comprises a spindle seat and a stop rack, wherein a first rotation portion is provided on each of two sides of the spindle seat along a width direction; the stop rack is detachably connected to the spindle seat, and a second rotation portion is provided on each of two sides of the stop rack along the width direction; swing arms, wherein two of the swing arms are rotatably disposed on two first rotation portions respectively; at least two slide rail arms, wherein the at least two slide rail arms are synchronously and rotatably disposed on two second rotation portions respectively, and each of the at least two slide rail arms is provided with a slide groove; and a connecting rod shaft, wherein one end of the connecting rod shaft is connected to a swing arm of the two swing arms, the other end of the connecting rod shaft is in a sliding fit with the slide groove, and the swing arm is driven to rotate through the slide groove and the connecting rod shaft in a rotation process of a respective one of the at least two slide rail arms.
2. The synchronization mechanism according to claim 1, wherein one of the stop rack or the spindle seat is provided with an overlapping groove, the other one of the stop rack or the spindle seat is provided with an inserting protrusion, and the inserting protrusion is inserted into the overlapping groove.
3. The synchronization mechanism according to claim 2, wherein the overlapping groove is disposed on the spindle seat, the overlapping groove passes through the spindle seat along a direction facing the stop rack, the overlapping groove is a stepped groove, and two first limit portions opposite to each other are formed at an opening of the stepped groove; and the inserting protrusion is disposed on the stop rack, the inserting protrusion comprises an inserting body extending along a length direction and second limit portions disposed at an end of the inserting body and extending along the width direction, a width of the inserting protrusion at the second limit portions is greater than a width of the overlapping groove at the two first limit portions, and the second limit portions are disposed on one side facing away from the opening of the overlapping groove.
4. The synchronization mechanism according to claim 1, wherein each of the two swing arms

- comprises a swing body and a connecting portion connected to one end of the swing body, the connecting portion is in a rotation fit with the first rotation portion, the swing body is provided with a swing hole, and the connecting rod shaft passes through and is secured within the swing hole.
5. The synchronization mechanism according to claim 4, wherein the swing body is provided with an arc-shaped groove passing through the swing body along a length direction, each of the two swing arms further comprises a cover plate covering the arc-shaped groove, the swing hole is formed by an enclosure between the cover plate and the arc-shaped groove, and a length of the cover plate is less than a length of the arc-shaped groove in the length direction.
6. The synchronization mechanism according to claim 4, wherein at least two connecting rod shafts are provided, the at least two connecting rod shafts are in one-to-one correspondence with the at least two slide rail arms, and each of the at least two connecting rod shafts has a rod body and a stop portion disposed at an end of the rod body, wherein the rod body passes through the swing hole, an outer diameter of the stop portion is greater than an outer diameter of the swing hole, and the stop portion is located on one side of the cover plate facing away from the respective one of the at least two slide rail arms.
7. The synchronization mechanism according to claim 6, wherein an end surface of the stop portion is fan-shaped and disposed around an outer periphery of the rod body, and two ends of the stop portion around an axis of the stop portion abut against the swing body and are located on two sides of the arc-shaped groove, respectively.
8. The synchronization mechanism according to claim 6, wherein the rod body comprises a first shaft and a second shaft, a diameter of the first shaft is greater than a diameter of the second shaft and greater than a width of the slide groove, the stop portion is disposed at an end of the first shaft facing away from the second shaft, the second shaft is slidably disposed in the slide groove, and the first shaft is in an interference fit with the swing hole or is bonded to the swing hole.
9. The synchronization mechanism according to claim 1, wherein a slide rail arm of the at least two slide rail arms rotates between an initial position and an end position, and the slide groove is bent downward gradually along a direction facing away from the base.
10. The synchronization mechanism according to claim 9, wherein each of the at least two slide rail arms is provided with a slide protrusion, the slide protrusion is configured to be in a sliding fit with a slide rail groove of a housing connecting rod, and the slide protrusion is provided with a notch for avoidance of the connecting rod shaft.
11. The synchronization mechanism according to claim 1, wherein the stop rack comprises a stop body and two hinged protrusions disposed on two sides of the stop body respectively, the second rotation portion comprises a hinged hole passing through a respective one of the two hinged protrusions along the length direction, two hinged blocks opposite to each other are provided on one end of each of the at least two slide rail arms, a hinged groove is formed between the two hinged blocks, a rotation hole passes through each of the two hinged blocks along the length direction, each of the two hinged protrusions is inserted into the hinged groove, and a driving shaft passes through the hinged hole and the rotation hole.
12. The synchronization mechanism according to claim 11, wherein each of the two hinged protrusions is provided with a first limit protrusion and a second limit protrusion, the first limit protrusion is arc-shaped and disposed around an outer periphery of the hinged hole, and the second limit protrusion is arc-shaped and disposed around an outer periphery of the rotation hole; and a slide rail arm of the at least two slide rail arms rotates between an initial position and an end position with respect to the stop rack, in response to the slide rail arm being at the initial position, one end of the first limit protrusion abuts against one end of the second limit protrusion, and in response to the slide rail arm being at the end position, the other end of the first limit protrusion abuts against the other end of the second limit protrusion.
13. The synchronization mechanism according to claim 11, wherein the driving shaft comprises a shaft body and a shaft limit portion, an outer diameter of the shaft body is less than an outer

diameter of the shaft limit portion, the shaft body passes through the two hinged blocks and the rotation hole, the shaft limit portion is located on a side of a respective one of the at least two slide rail arms facing the spindle seat, and the spindle seat is provided with an avoidance notch for accommodating the shaft limit portion.

14. The synchronization mechanism according to claim 13, further comprising two torsion assemblies, wherein each of the two torsion assemblies comprises two intermediate gears engaging with each other and two main gears disposed on outer sides of the two intermediate gears respectively, two slide rail arms on a same stop rack are connected to the two main gears respectively and rotate around axes of the two main gears, respectively, the two main gears and the two slide rail arms are relatively secured in a circumferential direction, the two intermediate gears are rotatably disposed on the stop rack, the two intermediate gears engage with the two main gears respectively, and the driving shaft passes through a gear hole of the two main gears.

15. The synchronization mechanism according to claim 14, wherein a main gear of the two main gears is provided with a limit protruding block located on a circumferential side of the gear hole, a hinged block of the two hinged blocks is provided with a limit groove disposed on a circumferential side of the rotation hole, and the limit protruding block is inserted into the limit groove; wherein two limit grooves are provided, one of the two limit grooves communicates with the rotation hole, the other one of the two limit grooves is spaced from the rotation hole, and two limit protruding blocks are provided and inserted into the two limit grooves, respectively.

16. A folding hinge, comprising door plates, housing connecting rods and a synchronization mechanism, wherein the synchronization mechanism comprises: a base, wherein the base comprises a spindle seat and a stop rack, wherein a first rotation portion is provided on each of two sides of the spindle seat along a width direction; the stop rack is detachably connected to the spindle seat, and a second rotation portion is provided on each of two sides of the stop rack along the width direction; swing arms, wherein two of the swing arms are rotatably disposed on two first rotation portions respectively; at least two slide rail arms, wherein the at least two slide rail arms are synchronously and rotatably disposed on two second rotation portions respectively, and each of the at least two slide rail arms is provided with a slide groove; and a connecting rod shaft, wherein one end of the connecting rod shaft is connected to a swing arm of the two swing arms, the other end of the connecting rod shaft is in a sliding fit with the slide groove, and the swing arm is driven to rotate through the slide groove and the connecting rod shaft in a rotation process of a respective one of the at least two slide rail arms; wherein two of the housing connecting rods are located on two sides of the base along the width direction respectively, are in a sliding fit with two slide rail arms respectively and are hinged with two swing arms respectively, and two of the door plates are located on the two sides of the base along the width direction respectively, are in a rotation fit with two of the housing connecting rods respectively and are in a rotation fit with two connecting rod shafts respectively.

17. The folding hinge according to claim 16, wherein a rotation groove is provided on a side of each of the housing connecting rods facing a respective one of the two swing arms, one first connecting hole is provided on each of opposite sidewalls of the rotation groove, a rotation protrusion is provided on a side of the respective one of the two swing arms facing the respective housing connecting rod, a second connecting hole passes through the rotation protrusion along a length direction, the rotation protrusion is inserted into the rotation groove, a connecting shaft passes through the second connecting hole, and two ends of the connecting shaft are located in two first connecting holes respectively.

18. The folding hinge according to claim 17, further comprising a median plate, wherein the spindle seat is provided with an overlapping groove, the stop rack is provided with an inserting protrusion, the inserting protrusion is inserted into the overlapping groove, the overlapping groove passes through the spindle seat along a thickness direction, and the median plate is soldered onto the spindle seat and covers the overlapping groove.

19. The folding hinge according to claim 18, wherein two first limit portions opposite to each other are formed at an opening of the stepped groove; and the inserting protrusion comprises an inserting body extending along a length direction and second limit portions disposed at an end of the inserting body and extending along the width direction, a width of the inserting protrusion at the second limit portions is greater than a width of the overlapping groove at the two first limit portions, and the second limit portions are disposed on one side facing away from the opening of the overlapping groove.

20. The folding hinge according to claim 19, wherein each of the two swing arms comprises a swing body and a connecting portion connected to one end of the swing body, the connecting portion is in a rotation fit with the first rotation portion, the swing body is provided with a swing hole, and the connecting rod shaft passes through and is secured within the swing hole.
